

Start here

Play

Rhythm

Tracks

Step settings

Layers

Studio

Export

Tips

■ Start here

Five steps, about two minutes. Everything is in tune, so you can't play a wrong note.

1. **Pick a rhythm.** Choose a tradition in the tabs, then a rhythm. The card below it says where the rhythm comes from.
2. **Press Play** (or the space bar). Set the tempo with *BPM*, or tap it in with *Tap* (key T).
3. **Tune it to your track.** Pick *Key* and *Scale* and press *Tune to key*.
4. **Shape the groove.** Turn *Density*, *Variation* and *Polymeter*, or press *Suggest* for three variations.
5. **Keep it.** Store it in a *Scene*, or export the loop as WAV and MIDI.

Try: hold *Drop* (key D) for two bars and let go. The drums fall away while the bells keep time, then everything slams back in.

■ Play

TRANSPORT

Play · Refresh!

Play starts and stops. **Refresh!** starts a clean jam: everything back to how Ngoma first opens, while tempo, key and scenes stay.

Refresh! resets rhythm, kit, groove, Pad, Lead, Magic and all effects. ← brings back what you had.

Drop · Drums off

Hold *Drop* (key D) for a breakdown: the drums drop out, bells, clave and shaker keep time. *Drums off* silences all drums until you press it again (key M).

← →

Step back and forth through your last 30 versions, like a browser.

Every randomize, new variant, Rnd, kit roll, reset and scene is kept as a version, so you can roll freely and return to the one that grooved.

Key · Scale · Tune to key

The key and scale of your track. *Tune to key* tunes the drums: low drums to the root, tumba and bongo to the fifth, djembe to the minor seventh. *Lock to scale* makes the Tune knobs land only on notes of the scale.

Chords · Per

One chord scheme for the whole piece. The pad, Worp, Resonate and the tuned drums follow it. *Per* sets how long each chord lasts.

Drone (the default) is Ngoma as it was: no changes. *Auto* picks a scheme that suits the tradition: Afro-Cuban 1·4·5·4, Caribbean 1·4, Middle Eastern and Southern Europe 1·7·6·5; North Indian stays on the drone. *Wander* finds its own way and comes home now and then. Every chord is built from your scale, so whatever you play on top still fits; in a pentatonic scale many chords have no third and show as sus4. The tuned drums move with the chord and keep their place in it (the root drum lands on the new root, the fifth on the new fifth); shakers, triangle, tambourine and claps stay. Switch that off with *Drums too*. The scope at the top shows the chord now and the next one. Export a loop at least as long as the scheme to get all of it.

MIDI	Play along from a MIDI keyboard. Click <i>MIDI: connect</i> once; after that Ngoma connects by itself. <i>Lead</i> or <i>Pad</i> picks what your keys play (the Worp pad; when the one you pick is off, the other plays). <i>Tight</i> makes every key wait for the next sixteenth while Ngoma plays, so what you play lands on the grid; a key just after the line plays at once.
Volume	Master volume for listening, at the end of the scenes line. Double-click for full. Exports are always at -16 LUFS, whatever it says.

SCENES AND MORPH

Scenes 1 to 8	Eight slots that keep everything you hear. Click an empty slot to store, a stored one to play it on the next bar. A scene keeps rhythm, kit, groove, pad, effects and chords, not tempo or key. Shift-click or right-click stores over a slot, Alt-click clears it, keys 1 to 8 play them. Build an intro, a groove and a breakdown, then move between them.
Morph	Lets a scene grow out of what plays now, over 1 to 32 bars. <i>Off</i> switches on the next bar. The knobs glide, the notes change one step at a time with the strong beats last, and the pad and anything that can't glide change halfway.

Try: store your groove in 1, mute two tracks and store that in 2. Set Morph to 8 and click 1 and 2 in turn.

■ Rhythm

CHOOSING A RHYTHM

Traditions and rhythms	Tabs for ten traditions, each with rhythms built from the parts each drum plays there. <i>Hypno</i> and <i>Euclidean</i> are style sketches, not traditions. <i>Origins</i> at the top tells where each rhythm comes from, what it means in its own tradition and which sources it is based on. Every rhythm card links to its own sources.
Back to the written rhythm	Puts the rhythm back exactly as written, with the default groove. Your sound, effects, tempo, key and locked tracks stay. The groove settings it resets: density, variation, swing and feel, humanize, life, polymeter, dialogue, fill, mutes and step settings. Clicking a rhythm chip again only restores its notes and keeps your settings, so you can keep playing.
Anchor	The timeline of the tradition, such as the clave or the bell. Suggest and Randomize leave it alone. Clave or bell in Afro-Cuban music, bell and dununba in West Africa, surdo and agogô in Brazil, the darbuka rhythm in the Middle East, the dembow on cajón, the conga figure in amapiano, the bayan of the theka in North Indian taal. Euclidean rhythms have no anchor: lock a track to make it one. Rnd on the anchor itself still works.

SHAPING THE GROOVE

Density	How many strokes play. 60% is the rhythm as written; lower thins out soft strokes first, higher adds ghost notes.
Variation	Changes that come and go, the way players vary. <i>Hypnotic</i> changes a note or two every fourth cycle, <i>Loose</i> varies more freely. The home loop always comes back. The timeline never moves, and the anchor only gets soft grace notes. The other drums change where players change: on the pickup into a beat and towards the end of the phrase, not on the downbeat. Accents stay; soft strokes drop out, turn into another stroke or get a grace note, never right against another stroke. Low drums vary least.

Swing and feel	Delays the offbeats on tracks with Sw on. 58% is the default; 50% is straight, 66% a full triplet. Click Sw again for half. <i>Swing 16th</i> counts like an MPC, in the range Roger Linn describes (54% loosens a straight beat, 62% swings loosely around 90 BPM). <i>Samba</i> plays the third and fourth sixteenth early and the first slightly late, as measured in Brazilian samba (Naveda et al. 2011). <i>Jembe (Mali)</i> uses the short-medium-long subdivision measured in Malian jembe music (Polak 2010); with a template, 50% is the measured feel. When a swung and a straight track land on the same step, hits that would end up 8 to 35 ms apart (where the ear hears a sloppy flam) are pulled together. Real flams and rolls stay.
Humanize	Small, repeatable differences in timing and velocity on every hit.
Life	The drums you hear most swell and ease over a few bars, and a thin high track adds a soft three-against-four ghost line. O is the rhythm exactly as written.
Polymeter	Gives some tracks an odd length, so they drift against the bar. Low drums and clave stay anchored longest. Shorter tracks cut the loop, longer ones keep cycling it. The orange badge on a track shows the length it plays. Between 25 and 45% grooves best; above 80% the groove loses its footing, good for a break.
Dialogue	Two players talk: a lead drum plays a short call at the end of one bar, another drum answers at the end of the next. Bongo, djembe, darbuka or tabla lead, like the quinto over the other congas; the answer comes preferably from another register, with the same rhythm on its own strokes. While a player talks it leaves its own part out. The timeline, the anchor, shaker and bells never take part. Higher means more notes and more conversations.
TRYING SOMETHING NEW	
Suggest	Makes 40 variations, scores them and offers the best three as A, B and C. Click one to hear it. The score looks at syncopation (a medium amount grooves best, as Witek et al. 2014 found), whether the density is in the range of the presets, whether accents fight for the same register on the same step, and whether the loop still repeats itself. The three are chosen so they really differ.
★ Keep	Keeps the notes of this pattern as a favourite above the tracks. A favourite loads only its notes; × removes it.
Surprise • New variant	<i>Surprise</i> gives a new take on the rhythm. <i>New variant</i> makes one to three small changes per track; accents stay.
Randomize	New steps, pitch, velocity or all, on the tracks that play. Locked tracks never change. It is not a coin toss: it weighs the metre, what each register usually does (low drums on strong beats and pickups, mid drums syncopated, high parts on the offbeats) and what the other tracks already play, and fills the gaps instead of doubling them.
Changes land on the bar	While Ngoma plays, a new rhythm or variation waits for the next bar line, so the groove never breaks. Knobs and sounds change at once.
KIT	
Roll kit • Reset kit	<i>Roll kit</i> gives every unlocked drum a new tuning (inside the scale), decay, pan and tone. <i>Dice amount</i> sets how far. <i>Reset kit</i> brings back the default kit.

■ Tracks

Tracks are grouped low, mid and high, each with its own colour: green, ochre and blue. *Only playing* hides the tracks this rhythm does not use.

READING A TRACK

Steps	Click for a hit, again for the next stroke, once more to clear. Shift-click or right-click toggles accent and soft. The letter shows the stroke and the shade tells them apart: full colour is open or bass, dark is mute or tone, light is slap or ka. A dashed outline marks a variation hit, a dot a step with settings.
Light	Lights up on every hit. Click it to mute the track.
M·S·bin	Mute, solo, and clear every note of the track (← brings them back).

SOUND OF A TRACK

Level·Tune·Decay·Pan	Volume, pitch in semitones, length of the ring and stereo position. Level goes past 100% into boost, up to 160%, and the ring turns red: for a surdo that has to work as a kick. Open and muted strokes sound on the note under the track name; slaps ring a third or fourth higher, always on a note of your scale.
Push	Moves the track up to 20 ms ahead of or behind the grid: a shaker slightly behind, a clave slightly ahead.
Var·~·Drift	<i>Var</i> makes every hit a little different, as real drums do. <i>~</i> adds a slow wander of pitch, colour and level over a few bars. Drift is off by default and goes up to about 25 cents. While it plays, the bar under <i>~</i> sways and the note under the name shows the offset, like B1 +13¢.
Rev·Dly	How much of the track goes to the reverb and the delay.
Dice	A new sound for this drum only: tuning, decay, colour, variation, pan and push. The rhythm stays.

TRACK OPTIONS

Length·Sw·Lock·Rnd·E	Steps in the track (2 to 32), swing on, keep this track when you change rhythm, a new part for this track only, and the Euclidean generator.
Choke	A new stroke stops what is still ringing, as a hand damps the skin. <i>A</i> and <i>B</i> let tracks in a group cut each other. For example conga and tumba played by one player, where a mute on one stops the open tone of the other. Flams and rolls are left alone.
Track controls	Opens or closes the knob row of every track. The round arrow next to a name folds one track.

■ Step settings

Pick an edit mode above the tracks, then click steps to cycle through values. Shift-click goes back, right-click resets.

Chance	100, 75, 50, 25 or 10% that the step plays each time.
Condition	<i>1:2</i> plays on the first of every two cycles, <i>3:4</i> on the third of four. <i>fill</i> plays only in a fill, <i>!fill</i> only outside one.
Flam / roll	<i>F</i> adds a soft grace stroke just before the hit; <i>R2</i> to <i>R4</i> make a roll of two to four strokes.
Pitch·Decay	Tune or shorten just this one step, for melodic figures on conga, udu or talking drum.
Fill	Hold Fill for a fill, or let <i>Auto fill</i> play one every 4, 8 or 16 bars.

■ Layers

What plays with the drums, always in your key, scale and chords. Each layer has its own On switch.

Pad

A soft layer under the drums. *Worp* throws a new patch from Ngoma's sister synth; *Classic* has six pads of its own.

Worp: every throw is a patch of up to four sources with slow modulation that keeps re-patching itself. *New* throws another, the numbered buttons mute its layers, *Open in Worp* opens Worp over Ngoma while Ngoma keeps playing (*Ngoma* or *Esc* brings you back). Whatever you pick or change there plays here at once; ← undoes a new sound.

Classic types: *Bed*, one chord per bar. *Loops*, five tone fragments on their own lengths, like a Buddha Machine. *Tides*, a four-voice chord whose voices swell on their own cycles. *Echoes*, two short phrases into a darkening tape echo. *Seasons* and *Cloud*, a slow chord walk with shared notes held, as a warm pad or a cloud of grains.

Knobs: *Level* (0 is really off), *Move* (how far the filter breathes, or how deep a Worp gate dips), *Filter* (low-pass to the left, high-pass to the right), *Reverb* and *Echo*, and *Pitch* (up a fourth, a fifth or an octave). Off at the first start.

Lead

A struck or plucked melody from Worp, in time and with the swing of the drums. Off by default.

Every lead has one of nine ways of making its line, shown next to its code: *Bounce* (after Dillon Bastar's Bouncy Notes), *Anchor* (the rhythm of the tradition's timeline), *Triplets*, *Loop*, *Phase* (after Steve Reich), *Modules* (after Terry Riley's In C), *Additive* (after Philip Glass), *Isorhythm* and *Pulse* (after Music for 18 Musicians). The sounds: marimba, vibraphone, kalimba, balafon, soft electric piano, reed, celesta or music box.

Knobs: *Level*, *Complexity* (left repeats longer and stays on the grid, right moves on sooner and bounces freely), *Filter*, *Reverb*, *Echo* and *Pitch*. *New* throws another lead, *Lock* keeps the line and changes only the sound.

Your own line: *Rec 2 bars* in Worp records one (a count-in, then two bars, quantized with the drums' swing). **Mine | Worp** switches between your line and Worp's.

Magic

Seven characters that work on the sound, not the rhythm. *Amount* says how far; everything stays in tune and in the export.

Breath: gentle life in timing and dynamics. **Resonate:** sympathetic strings on your key that ring with every stroke; *Chord* picks their notes (and with Chords on they follow the scheme). **Comb:** a tuned comb that moves to another note every two bars. **Air:** harmonics and sheen on top. **Formant:** two vowel filters, as if the kit speaks. **Cosmos:** a drifting echo after the SOMA Cosmos, two echo lines that drift apart and meet again. **Ghosts:** a tiny grain of a hit comes back a few steps later, quiet and far left or right.

Decay sets how long Resonate rings and Comb and Cosmos repeat. **Every** sends only every 2nd to 8th hit into Magic; the rest stays dry. **Echo** gives Magic its own dub echo. With Ghosts, **Chance** sets how often and **Hard** how present.

■ Studio

The units work like hardware: drag a knob up or down, hold Shift for fine moves, double-click for the default. The light by a name shows whether the unit is on.

SPACE

Reverb

Room, Plate, Hall, Cave and Spring, with length, pre-delay, tone and return. The low end stays out.

Built from a generated stereo impulse response: early reflections, density that builds up and highs that die faster than lows. *Spring* is a two-spring tank as in dub: every pass comes back as a small metallic chirp. Everything below 250 Hz is cut before the reverb. *Every* works as in Magic.

Delay	Synced to the tempo, 1/16 to 1/2. <i>Clean</i> or <i>Tape echo</i>; <i>Send</i> sets how much of the whole kit goes in. Clean plays ping-pong, stereo or mono. Tape echo is modelled after the Roland RE-201 Space Echo: three heads (<i>Heads</i>), a warm tone, wow and flutter, and saturation in the feedback, so high Feedback sings instead of blowing up. <i>Animate</i> lets feedback and tone drift in a cycle as long as the loop, so exports stay seamless. Below 300 Hz is cut.
tone	
Low • Mid • High	A simple master EQ. High starts at -2 dB, where long sessions tire the ear first. A low shelf at 150 Hz, a bell at 1.2 kHz where the skin tones sit and a high shelf at 4 kHz.
Filter: Clean • Ladder • MS • SEM	A low-pass over the whole mix, in four classic characters. <i>Cutoff</i>, <i>Reso</i> and <i>Drive</i> set it. Clean: transparent. Ladder: after the Moog ladder, 24 dB, round and warm. MS: after the Korg MS-20, 12 dB, keeps the bass and bites, screams at high resonance. SEM: after the Oberheim SEM, 12 dB, soft and open. Drive goes up to +18 dB with the level compensated, so louder does not fool your ear.
Move: Env • LFO • Rate	Makes the filter move: <i>Env</i> opens it on every hit, the <i>LFO</i> sweeps it in time with the loop. <i>Rate</i> is once per loop, bar, half bar or beat, so the export stays seamless; <i>Sync</i> switches to a free speed in Hz. <i>Env from</i> picks what opens the filter: the whole mix, or one track as a sidechain. A dot on the Cutoff knob shows where the filter is right now. Ngoma starts with the filter breathing lightly.
Master • Width • Room	The finishing chain: glue, a light tape saturation, the lows in mono, a little width and air. Loudness stays about the same, so On and Off compare fairly. Width spreads the players over the stereo field (0% is mono). Room puts the drums in a small wooden room, early reflections only; it is not the reverb. Stems skip Room and Master.
Weight • Kit pitch	<i>Weight</i> adds body and saturation inside every drum (25% by default). <i>Kit pitch</i> moves every drum up or down in semitones.
Flat	EQ flat, filter open, no drive, kit pitch 0, Master off and Weight 10%.

Try: filter MS, cutoff around 800 Hz, Reso 50%, Env 60%. Then set *Env from* to the conga.

■ Export

A LOOP

Export loop	A seamless WAV of 1, 2, 4 or 8 bars, plus MIDI and the pattern. Every export lands at -16 LUFS, peaks at least 1 dB under full scale. Reverb and delay tails fold back to the start, so the loop sounds as if it has been running forever. With effects on you also get a dry WAV. <i>JSON</i> keeps the full pattern and settings.
Stems	Every playing track as its own WAV as well, aligned with the mix, so they add up to it.
Leave room for kick	Leaves low strokes on the beat out of the export, so your own kick stays alone there. You still hear everything while you play.

WHAT YOU JUST PLAYED

Capture	Keeps the last 1, 2, 4 or 8 bars exactly as you heard them, Magic accidents and knob moves included. Key C. Export renders the pattern afresh; Capture keeps the take. You get a WAV (cut on the bar line, seamless) and a MIDI file. Works while playing and right after Stop. <i>Drag take</i> drags the WAV straight into your DAW.
Take	Everything you played since Play, up to about 100 seconds, as WAV plus MIDI. Key Shift+C.

● Rec	Records a whole jam, up to 12 minutes, starting on the next bar line. You get four WAVs that start together: the mix as you heard it, and the drums, the pad and Worp on their own, dry. Silent parts are left out.
INTO YOUR DAW OR SAMPLER	
Drag WAV·Drag MIDI	Drag the loop straight onto a track in your DAW (Chrome).
Save to folder	Pick a folder once, such as your Ableton User Library, and Export and Capture save straight into it, no zip.
→ Blackbox	One click (key B) writes a WAV into your Blackbox folder, named by tempo, key and rhythm. While playing it keeps the last bars as you heard them; stopped it renders the loop. Names start with the tempo in three digits, so the Blackbox lists them by tempo: <i>110 Am SonClaveTumb 4b 01.wav</i> . 48 kHz, 24-bit; the Blackbox counts the beats from the length.
MIDI layout	<i>Drum rack</i> puts one instrument per pad from C1, <i>One pad per stroke</i> gives every stroke its own note, <i>General MIDI</i> suits GM kits.

■ Tips

- **Fold what you don't use.** The arrow on a panel folds it away. A new session opens them all again, so nothing gets lost.
- **Registers.** Low gives weight, mid speaks the language of the rhythm, high gives drive. Don't let all three play busy at once.
- **One change at a time.** Hypnotic grooves come from repetition with small shifts. Try Variation on Hypnotic around 30% and leave it.
- **Accents make the groove.** The bells accent the main beats and play the strokes between softer, as players do.
- **Clay and wood.** Udu, cajón, bendir and talking drum sit outside the traditional rhythms. Try *Clay & wood*, or add them with Rnd.
- **Euclidean.** 3 over 8 is the tresillo, 5 over 8 the cinquillo, 7 over 12 the West African bell, 4 over 9 the Turkish aksak (Toussaint).
- **Odd meters.** Samai 10/8, Aksak 9/8 and Rupak (7 beats) run over the 4/4 grid, so they drift across the bar like a polymeter.
- **Always at hand.** Scroll down and a slim bar at the top keeps Play, tempo, Capture, Drop and your scenes in reach.

Where the rhythms come from and the sources behind them: **Origins**, the first button in the top bar.